Statistical Analysis of Experimental Data

Homework 02

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https://github.com/igordt/SAED_homeworks/tree/main

Introduction

This homework concerns the Bertrand paradox: *in a circle of radius R an equilateral triangle is drawn. What is the probability that a random chord is longer than the side of the triangle?*

As demonstrated in the lecture, the answer depends on how a random chord is defined. Below, I present my own definition of a random chord, the distribution of chords generated using this method, and the resulting probability.

Definition of the random chord

A random chord is generated as follows:

1. two points P_1 and P_2 are chosen independently and uniformly distributed inside the disk of radius R;
2. the chord is the segment of the line through P_1 and P_2 delimited by the circle.

Whether the circumference belongs to the set of possible points or not does not matter: it has zero area, so the probability that a point lies exactly on it is zero.

Uniform points inside the disk

Uniformly distributed inside the disk means that the probability of finding a point in a region is proportional to its area. In polar coordinates (r, φ) the angle φ can be taken uniform in $[0, 2\pi)$. For the radius, the first naive choice would be $r = R\,U$, with U uniform in $[0, 1)$. With this choice every ring of equal thickness dr receives the same number of points, but the area of a thin ring between r and $r + dr$ is

$$dA = 2\pi r\, dr, \tag{1}$$

which grows with r. The density of points per unit area would then be

$$\frac{dN}{dA} \propto \frac{dr}{2\pi r\, dr} \propto \frac{1}{r}, \tag{2}$$

so the points would accumulate near the centre.

To have points uniform in area, the probability of being in a ring must be proportional to its area. The probability density of r and its cumulative distribution[1] are therefore

$$f(r) = \frac{2r}{R^2}, \qquad F(r) = \int_0^r f(r')\, dr' = \frac{r^2}{R^2}, \qquad 0 \le r \le R. \tag{3}$$

Using the inverse transformation:

$$U = F(r) = \frac{r^2}{R^2} \quad \Longrightarrow \quad r = R\sqrt{U}. \tag{4}$$

[1] The cumulative distribution function $F_X(x)$ of a random variable X is the function that gives the probability that X takes a value less than or equal to a specified value x.

Chord from the two points

The line through $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ is written in parametric form as

$$x(t) = x_1 + t\,\Delta x, \qquad y(t) = y_1 + t\,\Delta y, \qquad \Delta x = x_2 - x_1, \quad \Delta y = y_2 - y_1. \tag{5}$$

Imposing $x(t)^2 + y(t)^2 = R^2$ gives the second-degree equation

$$a\,t^2 + b\,t + c = 0, \qquad a = \Delta x^2 + \Delta y^2, \quad b = 2(x_1\Delta x + y_1\Delta y), \quad c = x_1^2 + y_1^2 - R^2. \tag{6}$$

Since P_1 is inside the circle, $c \le 0$ and the discriminant $b^2 - 4ac$ is always non-negative. The two solutions $t_\pm$ give the endpoints A and B of the chord, and its length is $L = |B - A|$.

The procedure is illustrated in Figure 1.

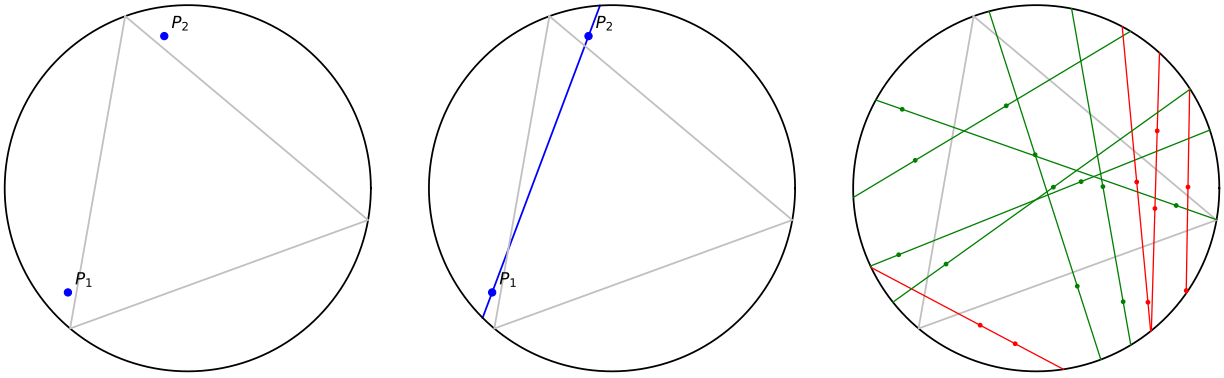


Figure 1: Definition of the random chord. Left: two random points P_1 and P_2 uniformly distributed inside the disk. Centre: the chord through the two points. Right: ten examples of random chords with their generating points; the chord is green if it is longer than the side of the inscribed triangle, red otherwise.

Distribution of the chords and probability

Following the procedure described above, $N = 1000$ random chords are generated. For each chord the length L is compared with the side of the triangle $\ell = R\sqrt{3}$, and the probability is estimated as the fraction of chords longer than the side:

$$P = \frac{n_{long}}{N}. \tag{7}$$

Each chord is an independent Bernoulli trial (two possible outcomes: longer or shorter than the side), so n_{long} follows a binomial distribution with variance $\mathrm{Var}(n_{long}) = NP(1-P)$. The uncertainty on the fraction $P = n_{long}/N$ is then[2]

$$\sigma_P = \sqrt{\frac{\mathrm{Var}(n_{long})}{N^2}} = \sqrt{\frac{P(1-P)}{N}}. \tag{8}$$

Out of $N = 1000$ chords, $n_{long} = 736$ are longer than the side of the triangle, which gives

$$P = 0.736 \pm 0.014. \tag{9}$$

The generated chords are shown in Figure 2. The chords cross the central region of the disk more often, and most of them are longer than the side of the triangle; the shorter chords are found near the border of the circle.

[2]For a random variable X and a constant a, $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$. With $X = n_{long}$ and $a = 1/N$ this gives $\mathrm{Var}(P) = \mathrm{Var}(n_{long})/N^2 = P(1-P)/N$.

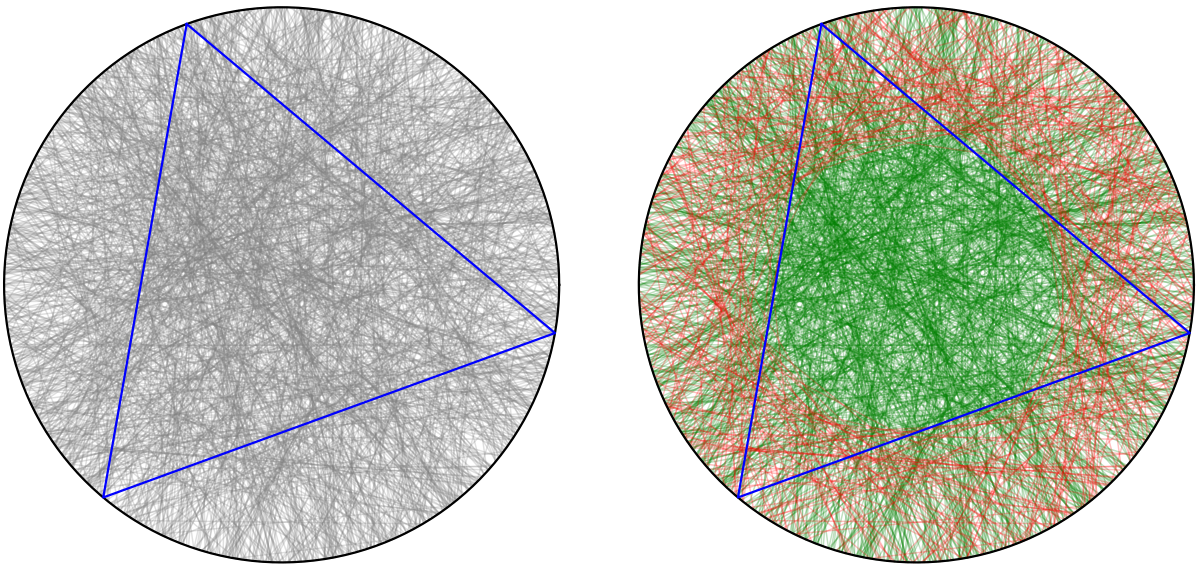


Figure 2: Left: the $N = 1000$ random chords generated with the procedure of Figure 1. Right: the same chords, in green if longer than the side of the inscribed triangle and in red otherwise.

Dependence of the estimate on the number of extractions

To study how the precision of the estimated probability improves with the number of extractions, the whole procedure is repeated for $N = 10^k$ chords, with $k = 2, \ldots, 7$. For each value of N an independent set of chords is generated, and the probability and its uncertainty are computed as in Eq. (8). The results are reported in Table 1.

N	P	σ_P
10^2	0.82000	0.03842
10^3	0.72800	0.01407
10^4	0.75080	0.00433
10^5	0.74746	0.00137
10^6	0.74680	0.00043
10^7	0.74701	0.00014

Table 1: Estimate of the probability $P(L > \ell)$ and its statistical uncertainty for different numbers of extractions N.

As N grows, the estimate of P stabilises and its uncertainty decreases. To quantify this behaviour, σ_P is plotted as a function of N in log-log scale (Figure 3) and fitted with the least-squares method (`scipy.optimize.curve_fit`) with a straight line

$$\log_{10} \sigma_P = \alpha \log_{10} N + c \qquad \Longleftrightarrow \qquad \sigma_P \propto N^{\alpha}. \tag{10}$$

The fit gives

$$\alpha_{fit} = -0.493 \pm 0.005, \tag{11}$$

compatible with the expected value $\alpha = -1/2$. The statistical uncertainty therefore decreases as $1/\sqrt{N}$: to gain one more significant digit, the number of extractions has to be increased by a factor of 100.

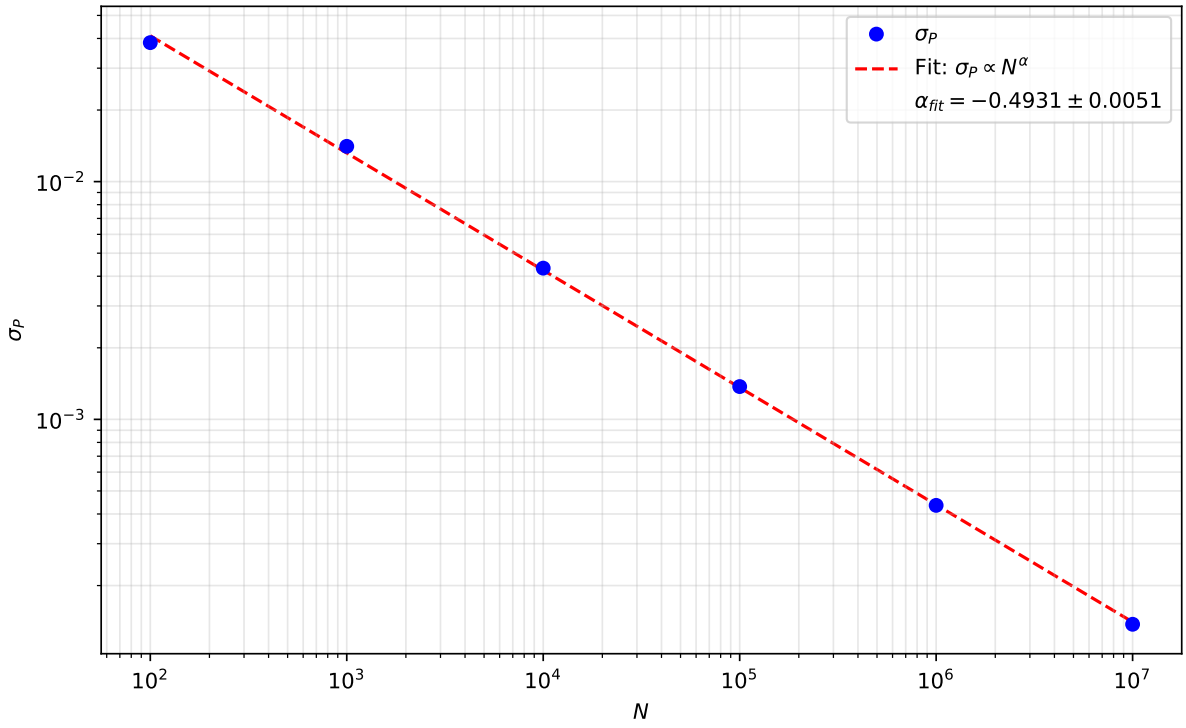


Figure 3: Statistical uncertainty σ_P as a function of the number of extractions N, in log-log scale. The dashed red line is the linear fit $\sigma_P \propto N^\alpha$.

Result

With the definition of random chord as the line through two points uniformly distributed inside the disk, the best estimate of the probability that a random chord is longer than the side of the inscribed equilateral triangle, obtained with $N = 10^7$ extractions, is

$$P = 0.74701 \pm 0.00014. \tag{12}$$

This value is different from all the three solutions presented in class (1/3, 1/2, 1/4), confirming that the answer to Bertrand's problem depends on the definition of random chord.